Submission STAT1-MID-SUB06

Question STAT1-MID-Q01
notation changes halfway:
1. x■ ≈ 25/5 ≈ 5.00
2. s^2 ≈ Σ(x■–x■)^2/(5–1) ≈ 14.00
marks, subscripts, and units are mixed up
result written as: mean=5.00; sample variance=14.00 (please interpret)

Question STAT1-MID-Q02
notation changes halfway:
1. P(D■∩D■) ≈ (5/16)((5-1)/(16-1))
2. ≈ 0.0833; the population shrinks after draw 1
marks, subscripts, and units are mixed up
result written as: 0.0833 (please interpret)

Question STAT1-MID-Q03
notation changes halfway:
1. SE ≈ √(p■(1–p■)/n) ≈ √(0.55(1–0.55)/40) ≈ 0.0787
2. CI ≈ 0.55±1.96(0.0787) ≈ (0.396, 0.704)
3. In repeated samples, the procedure captures the population proportion about 95% of the time.
marks, subscripts, and units are mixed up
result written as: (0.396, 0.704); repeated-sampling interpretation (please interpret)

Question STAT1-MID-Q04
notation changes halfway:
1. Increase sample size n to reduce standard error.
2. Reduce confidence level or population variability when justified.
3. Lower confidence changes coverage, so precision is not free.
marks, subscripts, and units are mixed up
result written as: increase n; lower confidence level or variability; note the coverage trade-off (please interpret)

Question STAT1-MID-Q05
notation changes halfway:
1. 0.03 < 0.05, so reject H■ under the stated rule.
2. Conclude evidence is inconsistent with H■; do not say the probability H■ is 3%.
marks, subscripts, and units are mixed up
result written as: reject H0; evidence against H0, not proof that H0 is false (please interpret)

Question STAT1-MID-Q06
notation changes halfway:
1. x■ ≈ 35/5 ≈ 7.00
2. s^2 ≈ Σ(x■–x■)^2/(5–1) ≈ 4.50

marks, subscripts, and units are mixed up

result written as: mean=7.00; sample variance=4.50 (please interpret)

Question STAT1-MID-Q07

notation changes halfway:

1. P(D■∩D■) ≈ (5/19)((5-1)/(19-1))
2. ≈ 0.0585; the population shrinks after draw 1

marks, subscripts, and units are mixed up

result written as: 0.0585 (please interpret)

Question STAT1-MID-Q08

notation changes halfway:

1. SE ≈ √(p■(1–p■)/n) ≈ √(0.35(1–0.35)/40) ≈ 0.0754
2. CI ≈ 0.35±1.96(0.0754) ≈ (0.202, 0.498)
3. In repeated samples, the procedure captures the population proportion about 95% of the time.

marks, subscripts, and units are mixed up

result written as: (0.202, 0.498); repeated-sampling interpretation (please interpret)